

```
mysql> show databases;
+-----+
| Database |
+-----+
| cg_mysql |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.01 sec)
```

```
mysql> use cg_mysql;
Database changed
mysql> show tables;
+-----+
| Tables_in_cg_mysql |
+-----+
| company_staff |
| employees |
| staff |
+-----+
3 rows in set (0.01 sec)
```

```
mysql> /* Constraints: */
mysql> /* They are rules applied to a table columns in MYSQL to ensure that only
valid and accurate data is stored in the database.
/*> In simple words Constraints = rules that control the data entered into a
table.
```

```
/*> Why do we use constraints?
/*> 1. maintain data integrity.
/*> 2. Prevent invalid data entry.
/*> 3. avoid duplicare records.
/*> 4. ensure relationships between tables.
/*> 5. Make the database reliable and consistent.*/
```

```
mysql> /* Integrity constraints;
/*> 1. not null
/*> 2. unique
/*> 3. Primary key
/*> 4. Composite key
/*> 5. foreign key
/*> 6. default
/*> 7. check */
```

```
mysql> /* not null: It ensures that a column cannot contain null(empty) values.
every row must have a value for that column.
```

```
/*> not null = doesn't allow null values. */
```

```
mysql> /* unique key: it ensures that all values in a column are different.
duplicate values are not allowed. */
```

```
mysql> /* primary key: it is a combination of unique key and not null */
```

```
mysql> /* Composite key: it is a table level constraint which is used to assign
```

multiple primary key columns in a table.

/\*> The values in composite key constraints are treated as a set and does not allow duplicate values.

/\*> Note: if even one value in the set is unique then it is treated as a new set even if it has duplicate values compared to other sets. \*/

mysql> /\* default: It automatically assigns a predefined value if no value is provided during insertion. \*/

mysql> /\* check: It ensures that values satisfy a specified condition before insertion or update. \*/

mysql> /\* foreign key: It creates a relation between two or more tables. It ensures that values in the child table must exist in the parent table.

/\*> the table where we assign foreign key is the child table and the tables to which it is linked to is the parent table. it is a tables level constraint. \*/

mysql> /\* Referencial Actions \*/

mysql> /\* Referential actions defines what happens to child table data when parent table data is updated or deleted.

/\*> commands: on delete cascade, on update cascade, on delete set null, on update set null \*/

```
mysql> create table products(  
-> product_id int primary key,  
-> product_name varchar(100),  
-> product_price decimal(10,2)  
-> );
```

Query OK, 0 rows affected (0.14 sec)

```
mysql> create table orders(  
-> order_id int primary key,  
-> customer_name varchar(30),  
-> product_id int,  
-> quantity int,  
-> foreign key(product_id),  
-> reference products  
-> );
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ', reference products

)' at line 6

mysql> show tables;

```
+-----+  
| Tables_in_cg_mysql |  
+-----+  
| company_staff      |  
| employees           |  
| products            |  
| staff               |  
+-----+
```

4 rows in set (0.01 sec)

mysql> drop table products;

Query OK, 0 rows affected (0.07 sec)

```
mysql> show tables;
+-----+
| Tables_in_cg_mysql |
+-----+
| company_staff      |
| employees           |
| staff              |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> create table departments(
-> department_id int primary key,
-> department_name varchar(20)
-> );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> create table doctors(
-> doctor_id int primary key,
-> doctor_name varchar(20),
-> doctor_salary decimal(10,2),
-> department_id int,
-> foreign key(department_id),
-> references departments(department_id),
-> on delete set null on update set null
-> );
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ', references departments(department\_id), on delete set null on update set null )' at line 6

```
mysql> create table doctors(
-> doctor_id int primary key,
-> doctor_name varchar(20),
-> doctor_salary decimal(10,2),
-> department_id int,
-> foreign key(department_id),
-> references departments(department_id),
-> on delete set null,
-> on update set null
-> );
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ', references departments(department\_id), on delete set null, on update set null ' at line 6

```
mysql> create table doctors(
-> doctor_id int primary key,
-> doctor_name varchar(20),
```

```

-> doctor_salary decimal(10,2),
-> department_id int,
-> foreign key(department_id)
-> references departments(department_id)
-> on update set null on delete set null
-> );

```

Query OK, 0 rows affected (0.14 sec)

```

mysql> insert into departments
values(01,'cardiology'),(02,'dermatology'),(03,'ent'),(04,'radiology'),(05,'neurology');

```

Query OK, 5 rows affected (0.07 sec)

Records: 5 Duplicates: 0 Warnings: 0

```

mysql> select * from departments;

```

| department_id | department_name |
|---------------|-----------------|
| 1             | cardiology      |
| 2             | dermatology     |
| 3             | ent             |
| 4             | radiology       |
| 5             | neurology       |

5 rows in set (0.00 sec)

```

mysql> insert into doctors
values(101,'Ramesh',50000.00,1),(102,'Suresh',48000.00,3),(103,'Madhan',70000.00,5),
(104,'Kabir',100000.00,4),(105,'kristine',80000.00,2);

```

Query OK, 5 rows affected (0.05 sec)

Records: 5 Duplicates: 0 Warnings: 0

```

mysql> select * from doctors;

```

| doctor_id | doctor_name | doctor_salary | department_id |
|-----------|-------------|---------------|---------------|
| 101       | Ramesh      | 50000.00      | 1             |
| 102       | Suresh      | 48000.00      | 3             |
| 103       | Madhan      | 70000.00      | 5             |
| 104       | Kabir       | 100000.00     | 4             |
| 105       | kristine    | 80000.00      | 2             |

5 rows in set (0.00 sec)

```

mysql> delete from departments where department_id = 3;

```

Query OK, 1 row affected (0.05 sec)

```

mysql> select * from departments;

```

| department_id | department_name |
|---------------|-----------------|
|---------------|-----------------|

|   |             |
|---|-------------|
| 1 | cardiology  |
| 2 | dermatology |
| 4 | radiology   |
| 5 | neurology   |

4 rows in set (0.00 sec)

mysql> select \* from doctors;

| doctor_id | doctor_name | doctor_salary | department_id |
|-----------|-------------|---------------|---------------|
| 101       | Ramesh      | 50000.00      | 1             |
| 102       | Suresh      | 48000.00      | NULL          |
| 103       | Madhan      | 70000.00      | 5             |
| 104       | Kabir       | 100000.00     | 4             |
| 105       | kristine    | 80000.00      | 2             |

5 rows in set (0.00 sec)

mysql> update departments set department\_id = 3 where department\_id = 4;

Query OK, 1 row affected (0.03 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select \* from departments;

| department_id | department_name |
|---------------|-----------------|
| 1             | cardiology      |
| 2             | dermatology     |
| 3             | radiology       |
| 5             | neurology       |

4 rows in set (0.00 sec)

mysql> select \* from doctors;

| doctor_id | doctor_name | doctor_salary | department_id |
|-----------|-------------|---------------|---------------|
| 101       | Ramesh      | 50000.00      | 1             |
| 102       | Suresh      | 48000.00      | NULL          |
| 103       | Madhan      | 70000.00      | 5             |
| 104       | Kabir       | 100000.00     | NULL          |
| 105       | kristine    | 80000.00      | 2             |

5 rows in set (0.00 sec)

mysql> notee